

AZURE CLOUD PROJECT

Azure Virtual Machine Extension Standardisation

Project Documentation – Requirements, Architecture, Cost, and Alternatives

Project Code: 24CC3046-P090 | Category: Operations | September 2026

1. Project Details

Azure Virtual Machine Extension Standardisation is an Azure operations and automation project that ensures Azure VMs have required monitoring and security components installed consistently.

Use cases: ensure monitoring/security agents exist; detect and remediate missing extensions.

Bottlenecks: extension failures can leave VMs partially configured; multiple agents can conflict and consume resources.

2. Architecture

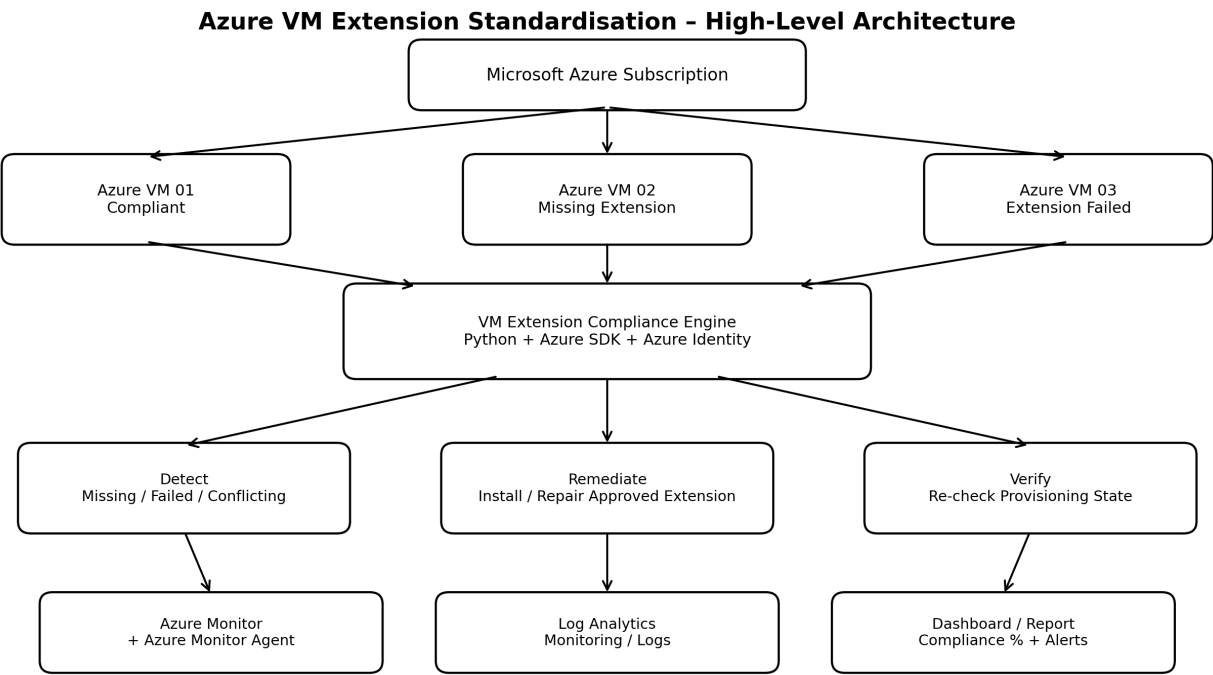


Figure 1. High-level architecture of the proposed system.

Flow: Azure Subscription → VMs → Extension Inspection → Compliance Engine → Detect → Remediate → Verify → Monitoring → Dashboard/Reports.

3. Project Requirements

Requirement	Use in our project	Required?	Cost / alternative
Azure Virtual Machines	Hosts the VMs whose extensions are checked/remediated.	Yes	PAYG-capable; student credit/free allowance. Local Hyper-V/VirtualBox only for development.
Virtual Network + Subnet	Provides VM networking.	Yes	Basic VNet/subnet has no separate charge. Local virtual switch outside Azure.
VM Extensions	Components being standardised and checked.	Yes	No separate extension-framework charge. PowerShell/Bash for local simulation.
Azure Monitor Agent	Standard monitoring agent checked on each VM.	Yes	Agent free; telemetry may cost. Prometheus Node Exporter locally.
Log Analytics	Stores/query logs for monitoring evidence.	Recommend ed	Consumption/PAYG. SQLite/CSV or OpenSearch/ELK locally.
Azure Monitor	Azure-native monitoring/metrics/alerts.	Recommend ed	Mixed pricing. Prometheus + Grafana locally.
Resource Graph	Efficient Azure inventory queries.	Recommend ed	Free. ARM/Compute SDK queries.

Requirement	Use in our project	Required?	Cost / alternative
Azure Automation	Scheduled scan/remediation.	Optional	Free allowance + possible PAYG. GitHub Actions/Task Scheduler.
Defender for Cloud	Security posture visibility.	Optional	Foundational CSPM free; advanced paid. OS security checks.
Key Vault	Secrets if required.	Optional	PAYG. Managed Identity is preferred alternative.
Storage Account	Azure report/history storage.	Optional	PAYG. Local CSV/PDF or GitHub artifacts.
Python + Azure SDK	Core automation/compliance logic.	Yes	Free. PowerShell/Azure CLI are alternatives.

4. Functional Requirements

ID	Requirement	Expected result
FR-01	Discover VMs	Automatic VM inventory.
FR-02	Inspect extensions	Extension names/states retrieved.
FR-03	Check compliance	Compliant/non-compliant/failed status.
FR-04	Detect missing/failed	Exact component identified.
FR-05	Remediate	Approved extension installed/repaired.
FR-06	Verify	Post-remediation state checked.
FR-07	Report	Compliance summary generated.
FR-08	Dashboard	Optional visual dashboard.
FR-09	Alerts	Optional failure notification.
FR-10	Scheduled scans	Optional recurring automation.

5. Free Tier, Student Credit, and PAYG Explained

Azure for Students: Microsoft currently lists USD 100 credit for eligible full-time university students for use within 12 months, with no credit card required, plus free monthly amounts for selected services.

General Azure Free Account: Microsoft currently lists USD 200 credit for up to 30 days plus selected free monthly service amounts, subject to eligibility.

Important: PAYG means a service can generate a charge when usage is outside an applicable free allowance or student credit. It does not mean the project must be expensive.

6. How Each Requirement Is Used in Our Project

Requirement	How we use it	Free / PAYG	Alternative
Azure VM	VM-01/02/03 are targets for extension compliance.	PAYG-capable; student/free credits may cover small lab.	Use 1–2 small Azure VMs; local VMs only for development.
Managed Disk	Stores VM OS.	Normally billable.	Minimum disk; local VHD outside Azure.
Azure Monitor Agent	Agent our checker verifies.	Agent free; telemetry may cost.	Prometheus Node Exporter locally.
Log Analytics	Monitoring/log evidence.	Consumption/PAYG.	SQLite/CSV or OpenSearch/ELK.
Azure Monitor	Metrics, logs, alerts.	Mixed pricing.	Prometheus + Grafana.

Requirement	How we use it	Free / PAYG	Alternative
Resource Graph	Azure resource discovery.	Free.	ARM/Compute SDK.
Azure Automation	Scheduled scans/remediation.	Free allowance + possible PAYG.	GitHub Actions / Task Scheduler.
Defender for Cloud	Security posture.	Foundational free; advanced paid.	OS security checks.
Key Vault	Secrets if needed.	PAYG.	Managed Identity.
Storage	Azure report/history.	PAYG.	Local CSV/PDF / GitHub artifacts.

7. Recommended Low-Cost Configuration

- Azure for Students if eligible.
- 1–2 small Azure VMs for the final demonstration.
- Azure Resource Graph for free inventory queries.
- Azure Monitor Agent with limited telemetry.
- Log Analytics only when required.
- No paid Defender plans for the basic project.
- Managed Identity instead of Key Vault.
- Local CSV/SQLite instead of Azure Storage.
- GitHub Actions/local scheduler instead of Azure Automation if not required.
- Stop/deallocate VMs when not testing and clean up unused resources.

8. Microsoft Azure Portal Setup

Portal resource	Configuration	Cost classification
Resource Group	rg-vm-extension-standardisation	No separate resource-group charge
Virtual Network	One VNet + subnet	Basic VNet/subnet – no separate charge
Virtual Machines	1–3 small VMs	PAYG-capable; use student credit/free allowance
VM Extensions	Required monitoring/security components	No separate extension-framework charge
Azure Monitor Agent	Install on target VMs	Agent free; telemetry can cost
Log Analytics	Create only if needed	Consumption/PAYG
Resource Graph	Use for inventory	Free
Automation	Optional scheduled jobs	Free allowance + possible PAYG
Defender	Foundational only	Free foundational; advanced paid
Cost Management	Monitor spend	Cost-control tool

9. Project Modules

Module	Responsibility	Output
M1	VM Discovery	VM inventory
M2	Extension Inspection	Extension inventory

Module	Responsibility	Output
M3	Compliance Engine	Compliance status
M4	Remediation	Remediation result
M5	Verification	Verified state
M6	Monitoring	Logs/metrics
M7	Dashboard	Dashboard
M8	Reporting	CSV/PDF
M9	Alerts	Notifications
M10	Testing	Evidence

10. Bottlenecks and Risk Controls

Bottleneck	Impact	Control
Extension failure	Partial configuration	Record, retry/escalate, review.
Conflicting agents	Resource/telemetry duplication	Standardise one monitoring path.
Permissions	Automation fails	Least-privilege RBAC.
Unexpected cost	Credit consumed	Small VMs, limited logs, cost monitoring.
Compatibility	Extension failure	Supported OS/images.
Network restrictions	Configuration failure	Verify outbound connectivity.

11. Deliverables and Acceptance Criteria

Deliverables: Azure test environment, Python source, extension standard, compliance/remediation module, reports, optional dashboard, test evidence, GitHub repository, presentation/viva documentation.

Acceptance: automatic VM discovery; correct detection; approved remediation; post-remediation verification; compliance report; failure demonstration; no hard-coded secrets; documented cost and alternatives.

12. Official Microsoft References

Azure for Students: <https://azure.microsoft.com/en-in/free/students/>

Azure Free Account / PAYG: <https://azure.microsoft.com/en-in/pricing/purchase-options/azure-account/>

Azure Free Services: <https://azure.microsoft.com/en-in/pricing/free-services/>

Azure Monitor Pricing: <https://azure.microsoft.com/en-in/pricing/details/monitor/>

Azure Monitor Agent: <https://learn.microsoft.com/en-us/azure/azure-monitor/agents/azure-monitor-agent-overview>

Azure Resource Graph: <https://learn.microsoft.com/en-us/azure/governance/resource-graph/overview>

Pricing note: Azure prices, quotas, regional pricing, and eligibility can change. Verify final costs in the Azure portal and Azure Pricing Calculator before deployment.